

## Membrane-Bound Cyclic AMP Phosphodiesterase in Chemotactically Responding Cells of *Dictyostelium discoideum*

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In cells of the cellular slime mold, *Dictyostelium discoideum*, that respond chemotactically to cyclic AMP, cell-bound cyclic AMP phosphodiesterase was studied under two conditions: in suspensions of living cells with external cyclic AMP as a substrate, and in the  $27000\times g$  sediment of cell homogenates. Under both conditions a drastic increase of cell-bound phosphodiesterase activity in the developmental phase preceding cell aggregation was found. In non-aggregating mutants this increase is smaller or absent. The particle-bound cyclic AMP phosphodiesterase has an optimum at pH 7.0–8.0. *In vivo* it is resistant to inhibition by external glutathione and to a macromolecular phosphodiesterase inhibitor from *Dictyostelium discoideum*; in particle fractions enzyme activity is partially sensitive to these inhibitors. A possible function of this enzyme in the chemotactic receptor system is discussed.

Development of the cellular slime mold, *Dictyostelium discoideum*, proceeds from single amoeboid cells to the formation of a multicellular organism by cell aggregation. Cells remain single during growth and differentiate later on within a preaggregation phase of several hours to aggregation-competent ones. These aggregation-competent cells are more sensitive to the chemotactic action of cyclic AMP than growth phase cells [1]; they are expected, therefore, to possess either a greater number of, or more sensitive cyclic-AMP receptors on their surface. Precise chemotactic orientation over a period of time requires continuous re-evaluation of the concentration gradient of the chemotactic agent by the responding cells. The following receptor mechanisms would fulfil such a function. (1) The receptor is a cyclic-AMP-binding compound without catalytic activity. Continuous functioning of the receptor system requires in this case either (a) high turnover of receptor molecules, or (b) rapid readjustment of the equilibrium of free to receptor-bound cyclic AMP, or (c) it requires enzymatic conversion of the bound cyclic AMP into a nonbinding product, *e.g.* by phosphodiesterase action. (2) The receptor itself acts as a phosphodiesterase. In this case the free energy change associated with hydrolysis of the phosphodiester ring would enable the cell to couple with hydrolysis

of cyclic AMP at the membrane an endergonic reaction, the product of which could mediate chemotactic motion. The  $\Delta G^\circ$  of cyclic AMP hydrolysis is  $-11.9$  kcal/mol (pH 7.3;  $25^\circ\text{C}$ ;  $\text{Mg}^{2+}$  equimoles to nucleotide) [2]. If possibility 1 or 2 is the case then a phosphodiesterase activity should be detectable on the cell surface, and this activity should be correlated with the responsiveness of the cells to cyclic AMP. In this paper we study the properties of such an enzyme with regard to its possible function as part of the chemotactic receptor system.

### METHODS

#### Cell Cultures

All *Dictyostelium discoideum* strains were cultivated in suspension on a gyratory shaker at  $23^\circ\text{C}$ . The ax-2 strain was grown, if not stated otherwise, in peptone-yeast extract medium supplemented with 1.8% maltose [3] up to a density of approx.  $10^7$  cells/ml. To induce differentiation to aggregation-competence, cells were washed 3 times and resuspended in 0.0167 M phosphate buffer pH 6.0. The suspension was adjusted to  $10^7$  cells/ml, and shaking continued. For growth on bacteria, *D. discoideum* cells were cultivated in a suspension of  $10^{10}$  cells/ml of *Escherichia coli* B/r or, in the case of mutant aggr50, of *Salmonella minnesota* R595 in phosphate buffer [4]. At the time of complete consumption of bacteria,

**Abbreviation.** Cyclic AMP, adenosine 3':5'-monophosphate.

**Enzymes.** Alkaline phosphatase (EC 3.1.3.1); adenosine deaminase (EC 3.5.4.4).

*D. discoideum* cells were washed 3 times and resuspended at a density of  $10^7$  cells/ml in phosphate buffer.

#### Assay in vivo of Cell-Bound Phosphodiesterase

Ax-2 cells suspended in phosphate buffer were washed 3 times, resuspended in phosphate to a final density of  $10^7$  cells/ml in a total volume of 300  $\mu$ l and rotated at 30 rev./min in small glass tubes. After preincubation with inhibitor (100 units/ml), the action was started with 50 nmol/ml of  $^3$ H-labeled cyclic AMP (25 Ci/mmol, Radiochemical Centre, Amersham, diluted to 7 Ci/mol) and stopped by adding 0.1 ml of 55% trichloroacetic acid. Cyclic AMP, 5'-AMP and adenosine were separated on Whatman No. 1 paper with 95% ethanol—1 M ammonium acetate (70:30, v/v) and counted in an Intertechnique SL 30 scintillation counter [5]. Glutaraldehyde-treated cells were prepared by incubation for 15 min at 4 °C in 0.01 M cacodylate buffer pH 7.2 containing 0.25% glutaraldehyde, washed 3 times and tested like living cells.

#### Soluble Extracellular Phosphodiesterase

Excretion of soluble phosphodiesterase was determined under the same conditions as cell-bound phosphodiesterase, except that no cyclic AMP was added and the supernatant of the cell suspension was harvested by centrifugation without addition of trichloroacetic acid. Phosphodiesterase activity was determined at 23 °C with alkaline phosphatase and adenosine deaminase in an UV photometric test [6] as described previously [7].

#### Particle-Bound Phosphodiesterase

Cells suspended in phosphate buffer were washed 3 times and resuspended at a density of  $4 \times 10^8$  to  $6 \times 10^8$  cells/ml in 0.01 M Tris-HCl buffer pH 7.4 containing 0.02 M  $\text{MgSO}_4$ . The cells were homogenized 3 min with glass beads of 0.5 mm diameter on a vibrator (D. Brück, personal communication), and the homogenate was successively centrifuged for 5 min at  $300 \times g$ , 10 min at  $1000 \times g$  and 20 min at  $27000 \times g$ . The  $27000 \times g$  sediment was washed 3 times in Tris-Mg buffer and used for assay of particle-bound phosphodiesterase. This membrane preparation was essentially free of nuclei but not of mitochondria. For crude particle preparations the washed cells were frozen, and after thawing the debris diluted into Tris-Mg buffer. The  $27000 \times g$  sediment obtained by 20-min centrifugation was washed 3 times.

Particle-bound phosphodiesterase was determined at 35 °C in 0.05 M Tris-HCl pH 7.4, containing 0.02 M  $\text{MgSO}_4$  and per ml about 100  $\mu$ g protein of the enzyme preparation and 68 nmol cyclic  $^3$ H-AMP (17 Ci/mol). The enzyme preparation was preincubated at 35 °C for 5 min, then cyclic AMP added up to a total

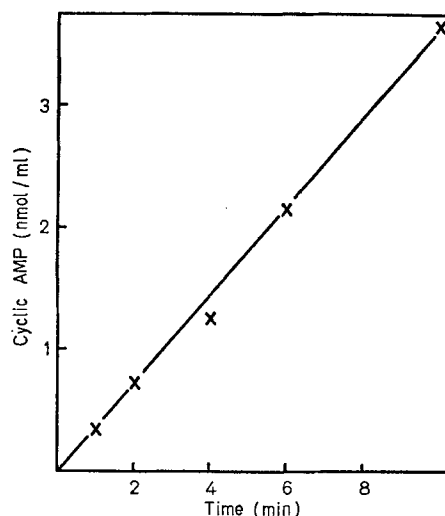


Fig.1. Hydrolysis of cyclic AMP by particle-bound phosphodiesterase as a function of time. Source of the particles:  $27000 \times g$  sediment of ax-2 cells, harvested 8 h after removal from growth medium, and homogenized with glass beads. The final concentration of particles was equivalent to 144  $\mu$ g protein per ml

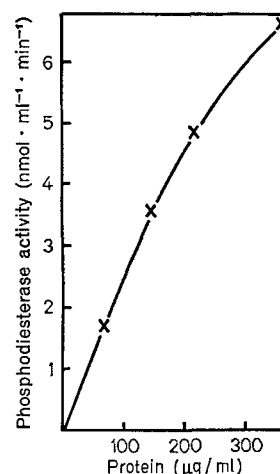


Fig.2. Reaction rate of particle-bound phosphodiesterase as a function of particle concentration under the conditions described in the text. The ordinate value of 6.8 indicates total hydrolysis of cyclic AMP within 10 min. Particles were prepared as described in legend to Fig.1

volume of 100  $\mu$ l, and after 10 min incubation in a shaking water bath the reaction stopped with 10  $\mu$ l of 55% trichloroacetic acid. Cyclic AMP and 5'-AMP were determined after chromatographic separation as above. Under these conditions, the reaction rate was constant over at least 10 min (Fig.1) and was practically a linear function of particle concentration up to a rate of 0.36 nmol cyclic AMP/min (Fig.2). All assays of particle-bound phosphodiesterase were done within these limits.

pH-dependence of particle-bound phosphodiesterase was measured in 0.08 M cacodylate buffer, pH 4.5–8.0 and 0.08 M Tris-HCl buffer, pH 6.5–9.0, both containing 0.02 M MgSO<sub>4</sub>.

#### “Soluble” Intracellular Phosphodiesterase

The 27000×*g* supernatant of cell homogenates was centrifuged 4 h at 100000×*g* and the supernatant of this centrifugations tested for soluble intracellular phosphodiesterase in the same photometric assay as for soluble extracellular phosphodiesterase. *D. discoideum* inhibitor was assayed as described previously [9], and 1 inhibitor unit defined as that quantity which, in a total volume of 300 µl, inhibits soluble extracellular phosphodiesterase with an activity of 2 nmol cyclic AMP/ml per min (at 35 °C) to one half [9]. The inhibitor was prepared from mutant ga86 by heating culture supernatants for 10 min at 80 °C. For inhibition studies, particle-bound phosphodiesterase was incubated with the crude inhibitor solution for 15 min at 35 °C in a total volume of 30 µl, and subsequently assayed. For interaction with living cell suspensions, the heated inhibitor was purified by acidification to pH 3.3, followed by ammonium sulfate precipitation and DEAE-cellulose chromatography.

Protein was determined according to Lowry *et al.* [11] with Lab-trol (Dade, Miami) as standard.

## RESULTS

### In vivo Hydrolysis of Cyclic AMP by Cell-Bound Phosphodiesterase

If a phosphodiesterase activity is associated with cyclic-AMP-receptor function in the chemotactic response of *D. discoideum* cells, a cyclic-AMP-hydrolyzing phosphodiesterase with the following properties would be expected: it should be an enzyme bound to the cell membrane with the catalytic site exposed to the extracellular space, it should be maximally active in aggregation-competent cells, and it should be insensitive to an extracellular phosphodiesterase inhibitor, which is produced during the preaggregation phase [7,12]. Consequently, suspensions of washed cells, harvested from axenic cultures of *D. discoideum* strain ax-2 [13], were assayed for cyclic AMP phosphodiesterase in presence of excess inhibitor. Although the inhibitor is known to almost completely inactivate soluble cyclic AMP phosphodiesterase excreted by *D. discoideum* cells [9], a phosphodiesterase activity resistant to the inhibitor was found in the cell suspensions. This activity strongly depended on the developmental stage of the cells.

When ax-2 cells are removed from their growth medium and washed, they undergo a series of changes which eventually after 7 to 9 h terminate in maximal aggregation-competence, as indicated by rapid cell

Table 1. Phosphodiesterase activity in living ax-2 cell suspensions

The data represent nmol cyclic AMP hydrolyzed from a total of 50 nmol by 10<sup>7</sup> cells suspended in 1 ml. Under these conditions, deviation from linearity caused less than 20% error up to a hydrolysis of 31 nmol cyclic AMP. The data in brackets exceed this limit. The two figures in column A are the results of two phosphodiesterase determinations from the same experiment

| Developmental stage | Phosphodiesterase activity in |                            |                                                              |
|---------------------|-------------------------------|----------------------------|--------------------------------------------------------------|
|                     | Cells incubated 15 min (A)    | Cells incubated 30 min (B) | Cells incubated 15 min, and supernatant incubated 15 min (C) |
| h                   | nmol                          | nmol                       | nmol                                                         |
| 1                   | 2.3<br>2.0                    | 4.7                        | 2.8                                                          |
| 3                   | 2.6<br>1.9                    | 7.0                        | 3.0                                                          |
| 5                   | 6.6<br>5.5                    | 14.4                       | 6.6                                                          |
| 7                   | 21.5<br>21.3                  | (41.8)                     | 30.8                                                         |
| 9                   | 32.5<br>26.2                  | (43.0)                     | 30.0                                                         |
| 11                  | 20.3<br>19.2                  | (34.3)                     | 21.2                                                         |

association due to chemotactic response and cell-to-cell adhesion. In the following, the developmental stage of a cell suspension is expressed in hours after removing cells from their growth medium. In all experiments cells were washed again immediately before use.

Table 1 (column A) and Fig. 4 show that the cyclic-AMP-hydrolyzing activity increased about 15-fold reaching a maximum at 9 h. To assure that the activity measured represents virtually cell-bound phosphodiesterase, aliquots of the suspensions were centrifuged after 15 min of incubation, and the almost cell-free supernatants incubated for another 15-min period. Only slow, if any continuation of cyclic AMP hydrolysis was observed in the supernatants (column C), compared with the noncentrifuged cell suspensions (column B; the 7-h sample being a possible exception).

Inhibitor was added to the cell suspensions in order to reliably eliminate the contribution of excreted phosphodiesterase to cyclic AMP conversion. But even in the absence of added inhibitor the extracellular enzyme activity remained too low within the first 20 or 40 min of incubation to contribute significantly (Fig. 3). Cell lysis as a possible source of discharged intracellular phosphodiesterase was also excluded by protein determinations in 8-h samples. Only 2% (0.02 mg/ml) of the cellular protein content (0.95 mg/ml) was found in the extracellular fluid after 60 min

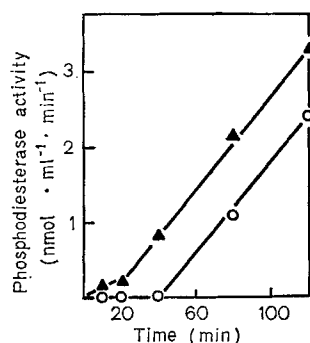


Fig. 3. Excretion of soluble extracellular phosphodiesterase in suspensions of  $10^7$  cells/ml washed ax-2 cells. Abscissa: time of harvest of the culture supernatant after washing the cells. O, 0-hour cells;  $\blacktriangle$ , 8-hour cells

of incubation. Because this figure included the proteins excreted by living cells, cell-loss was obviously less than 2%. Microscopic observations throughout incubation revealed healthy cells exhibiting pseudopodial activity.

After incubation with living 10-h cells, 98% of the cyclic  $[^3\text{H}]$ AMP label was found in the incubation medium, either as cyclic AMP or as 5'-AMP. When the same experiment was done with glutaraldehyde-treated cells, 30% of the label appeared in adenosine. Thus the 5'-AMP formed from external cyclic AMP is accessible only in these killed cells to intracellular phosphatase. It is improbable, therefore, that the cyclic AMP is taken up and split within living cells.

#### Increase of Particle-Bound Phosphodiesterase Correlated with Aggregation-Competence

Particle-bound phosphodiesterase was determined in the  $27\,000\times g$  sediment of cell homogenates. Similarly to the dynamics of cell-bound phosphodiesterase in the system *in vivo*, the specific activity of the particle-bound phosphodiesterase increased with cell development up to a peak at 9 h (Fig. 4).

Differentiation of growth-phase cells to aggregation-competent ones is inhibited in suspension cultures of the NC-4 strain of *D. discoideum* and its descendant ax-2 by a factor of low molecular weight (unpublished observation and [15]). Therefore, as in the experiments with living cells, ax-2 cells were washed in order to induce differentiation to aggregation-competence, and unwashed cells were used as a nondifferentiated control. The activity of particle-bound phosphodiesterase in bacteria-grown ax-2 cells followed their developmental competence: 9 h after consumption of bacteria, the particle-bound phosphodiesterase activity was 0.5 nmol cyclic AMP/mg protein per min in unwashed cells which did not differentiate, and 12.3 nmol cyclic AMP/mg protein

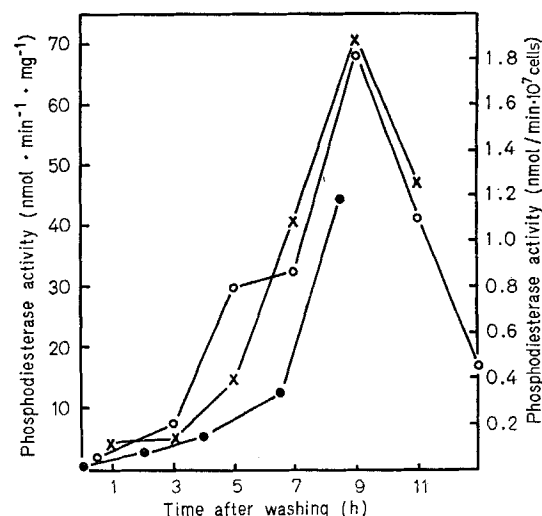


Fig. 4. Phosphodiesterase activity in particles, and in living cells of ax-2 as a function of developmental stage. Abscissa: hours after removing cells from growth medium. Left ordinate: specific phosphodiesterase activity in particles in nmol cyclic AMP hydrolyzed/min per mg protein, right ordinate: cell-bound phosphodiesterase activity in suspensions of  $10^7$  living cells per ml. O, Particle-bound phosphodiesterase in the  $27\,000\times g$  sediment of cell homogenates prepared with glass beads;  $\bullet$ , particle-bound phosphodiesterase activity in crude particles prepared by freezing and thawing.  $\times$ , cell-bound phosphodiesterase activity in suspensions of living cells. These data were calculated from Table 1 by correcting the values in column A for extracellular soluble phosphodiesterase: the activity registered in column C minus that in column A, divided by 2, was taken as an approximation of the mean activity of extracellular, soluble phosphodiesterase during the first 15-min period of incubation. This value was subtracted from the mean of the column A data

per min in aggregation-competent cells which had been washed at the time of complete consumption of bacteria.

#### Deficient Particle-Bound Phosphodiesterase in Non-Aggregating Mutants

The non-aggregating mutant aggr50 is deficient in inhibitor production, but not in extracellular, soluble phosphodiesterase [7]. This mutant did not show the increased activity of particle-bound phosphodiesterase characteristic of aggregation-competent wild-type cells (Table 2). Similarly, the assay *in vivo* of cell-bound phosphodiesterase in aggr50 5 h after consumption of bacteria, revealed an activity of not more than 0.06 nmol cyclic AMP/ $10^7$  cells per minute. Because the mutant cells possess the particle-bound phosphodiesterase activities typical for growth-phase cells, they are obviously deficient in the mechanism which is responsible for developmental regulation of this enzyme, but not deficient in its synthesis. The

Table 2. Particle-bound phosphodiesterase activity in the wild-type and in three lines of mutant *aggr50*

The data representing end of growth phase levels were obtained from cells harvested at the time of complete consumption of bacteria. 5 h later cells of the wild-type v-12/M2, and also those of the revertant *aggr50-3(rev)* are fully aggregation-competent. The activities were measured in crude particles prepared after freezing and thawing

| Strain No.                                                        | Phenotype                            | Cyclic AMP hydrolyzed |                         |
|-------------------------------------------------------------------|--------------------------------------|-----------------------|-------------------------|
|                                                                   |                                      | End of growth phase   | 5 h after end of growth |
| nmol $\times$ min <sup>-1</sup> $\times$ mg protein <sup>-1</sup> |                                      |                       |                         |
| v-12/M2                                                           | wild-type                            | 1.8                   | 23.8                    |
| aggr50/1                                                          | no aggregation                       | 4.1                   | 1.4                     |
| aggr50/2                                                          | no aggregation                       | 2.5                   | 2.7                     |
| aggr50/3(rev)                                                     | aggregation and sporophore formation | 2.4                   | 35.4                    |

Table 3. Activities of particle-bound phosphodiesterase in the wild-type and in aggregation-deficient mutants of *ax-2*

Cells grown on bacteria and washed at the end of growth were harvested 8 h later, and crude particles prepared after freezing and thawing

| Strain*                                                           | Cyclic AMP hydrolyzed |
|-------------------------------------------------------------------|-----------------------|
| nmol $\times$ mg protein <sup>-1</sup> $\times$ min <sup>-1</sup> |                       |
| <i>ax-2</i>                                                       | 39.7                  |
| <i>wag-1</i>                                                      | 0.8                   |
| <i>wag-2</i>                                                      | 0.6                   |
| <i>wag-3</i>                                                      | 6.0                   |
| <i>wag-4</i>                                                      | 3.0                   |
| <i>wag-5</i>                                                      | 16.0                  |
| <i>wag-6</i>                                                      | 8.0                   |
| <i>wag-7</i>                                                      | 2.4                   |
| <i>wag-8</i>                                                      | 9.5                   |
| <i>wag-9</i>                                                      | 17.0                  |
| <i>wag-10</i>                                                     | 5.0                   |
| <i>wag-11</i>                                                     | 7.5                   |

\* The strains are a gift from Dr J. Ashworth, Leicester.

*aggr50-3 (rev)* clone of *aggr50*, a spontaneous revertant to the aggregation-positive phenotype, had also regained the capacity to increase the specific activity of particle-bound phosphodiesterase during the pre-aggregation phase (Table 2).

Particle-bound phosphodiesterase determinations in a series of other aggregation-deficient mutants (Table 3) revealed reduced activities of this enzyme in all mutants tested, ranging from activities as low as in growth-phase cells of the wild-type, up to activities reaching nearly one half of the activities found in the wild-type during aggregation-competence.

#### pH-Dependence of the Particle-Bound Cyclic AMP Phosphodiesterase

The enzyme has, in the membrane-bound state, a broad optimum between pH 7.0 and 8.5 (Fig. 5), which

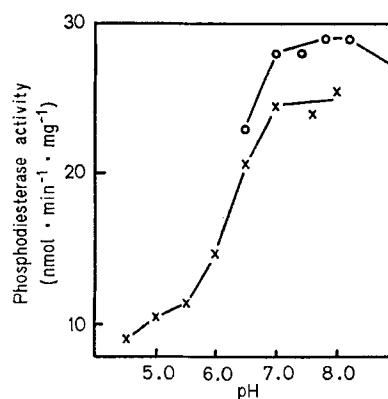


Fig. 5. pH-dependence of particle-bound phosphodiesterase. Source of the particles was the same as for Fig. 1. O, Tris-HCl buffer; X, cacodylate buffer

Table 4. Inhibitor action on cyclic AMP phosphodiesterases of *D. discoideum*

The concentration of *D. discoideum* inhibitor was equivalent to 17 units/ml, except in the experiment *in vivo*, where it was 100 units/ml. Source of extracellular, soluble phosphodiesterase was culture supernatant of the inhibitor-deficient mutant *aggr50*; source of the others were 8-hour cells of *ax-2*. Particle-bound and intracellular, soluble phosphodiesterases were prepared from cells homogenized with glass beads, and represent the activities in the washed 27000 $\times g$  sediment and in the 100000 $\times g$  supernatant, respectively

| Phospho-<br>diesterase                               | Activity<br>before<br>inhibition            | Activity left in presence of inhibitor |                      |                       |
|------------------------------------------------------|---------------------------------------------|----------------------------------------|----------------------|-----------------------|
|                                                      |                                             | D. dis-<br>coideum<br>inhibitor        | glutathione<br>10 mM | cyclic<br>IMP<br>5 mM |
|                                                      | nmol × ml <sup>-1</sup> × min <sup>-1</sup> | %                                      | %                    | %                     |
| Extracellular,<br>soluble                            | 7                                           | 3                                      | 0                    | < 5                   |
| Intracellular,<br>soluble                            | 8                                           | 15                                     | 0                    | not tested            |
| Particle-bound                                       | 3                                           | 40—70                                  | 20                   | 5                     |
| Cell-bound <i>in vivo</i><br>(10 <sup>7</sup> cells) | 1                                           | 100                                    | 85                   | <10                   |

resembles that of the extracellular, soluble phosphodiesterase [9]. It differs from cellular phosphodiesterase of presumably lysosomal origin, which is not cyclic-nucleotide-specific, is soluble and has an optimum of pH 4.5 [9].

#### Inhibitor Sensitivity of Cyclic AMP Phosphodiesterase

The cyclic-AMP-phosphodiesterase activities found in *D. discoideum* were compared with respect to inhibition by cyclic IMP, glutathione and the phosphodiesterase inhibitor of *D. discoideum* (Table 4). Cyclic IMP inhibited all the enzymes tested, whereas the *D. discoideum* inhibitor and glutathione inhibited

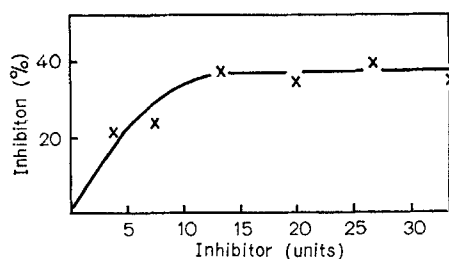


Fig. 6. Inhibition of particle-bound phosphodiesterase by *D. discoideum* inhibitor. Source of the particles was the same as for Fig. 1. The phosphodiesterase in the incubation mixtures had an activity of 2.7 nmol cyclic AMP/ml per min in the absence of inhibitor

only the soluble intracellular and extracellular phosphodiesterases completely. This suggests that both these activities represent identical enzymes. The cell-bound phosphodiesterase of living cells is almost resistant to both *D. discoideum* inhibitor and glutathione. In the cell-free system, however, the particle-bound phosphodiesterase is partially inhibited by these compounds. Increasing concentrations of *D. discoideum* inhibitor inhibited up to a plateau (Fig. 6), the level of which varied between 30 and 60% inhibition with different particle preparations. Even a concentration of 600 units of purified inhibitor per ml did not enhance inhibition beyond the plateau level. According to these results, the particle-bound phosphodiesterase is either heterogeneous; consisting of an inhibitor-sensitive and a resistant enzyme, the latter being identical with the cell-bound phosphodiesterase demonstrated *in vivo*; or it is one enzyme which is protected against inhibitor action in the intact cell, and becomes partially exposed during cell homogenization and fractionation. The finding that *D. discoideum* inhibitor and glutathione inhibit the particle-bound phosphodiesterase to different extents (Table 4) can be most easily explained in accepting the second alternative; in this case the portion of enzyme molecules with exposed glutathione-sensitive sites would be greater than that with exposed binding sites for *D. discoideum* inhibitor in a given particle preparation.

#### DISCUSSION

The experiments started with the question if a cyclic-AMP-phosphodiesterase activity is associated with chemotactic receptor function in aggregation-competent cells of *Dictyostelium discoideum*. A cell-bound phosphodiesterase has been found which is exposed in living cells to external cyclic AMP and is presumably localized therefore on the cell surface. This, as well as its developmental dynamics and inhibitor resistance, would predispose this enzyme for a function in chemotactic response as described in the Introduction. The following properties of the

cell-bound phosphodiesterase are different from that of the soluble, extracellular phosphodiesterase. Whereas the latter is produced mainly during growth, and is inactivated by a macromolecular inhibitor early in the stationary phase [7, 12], the activity of the cell-bound phosphodiesterase is low at the end of growth and increases during the stationary phase up to a peak at 9 h after removal of ax-2 cells from the growth medium. Furthermore, the cell-bound phosphodiesterase is not significantly inactivated *in vivo* by addition of two inhibitors of the soluble extracellular phosphodiesterase, namely glutathione and the *D. discoideum* inhibitor (Table 4). Despite these differences, it is possible that the surface-bound and the soluble extracellular phosphodiesterases are modifications of the same enzyme, because the inhibitor resistance may follow from the protection of sensitive sites due to incorporation of the enzyme into the membrane. The activities of cell-bound phosphodiesterase measured in washed and unwashed wild-type cells, as well as those found in the aggregateless mutants and in a revertant, indicate that the increase of activity is associated with the differentiation of cells from the growth-phase state to aggregation-competence; and not only with the shift from growth to stationary phase conditions. This increase of cell-bound phosphodiesterase activity parallels the rise of chemotactic sensitivity for cyclic AMP in *D. discoideum* cells during the shift from growth to aggregation [1].

Neither of these results convincingly proves that the surface phosphodiesterase is in fact a component of the chemotactic receptor system. Nevertheless, it may be calculated how many cyclic AMP counts a cell could maximally take per time, if it counted the external cyclic AMP molecules hydrolyzed by cell-bound phosphodiesterase. It follows from Fig. 4 that the aggregation-competent 9-h cells converted  $1.2 \times 10^8$  molecules cyclic AMP/cell per minute at saturating concentrations of the substrate. Evaluation of a chemotactic gradient would be possible only at concentrations that do not completely saturate the receptors. Therefore, the efficient number of counts a cell could take to measure a concentration gradient along its surface would be below the limit of  $1.2 \times 10^8$  counts per minute, if the hydrolysis of cyclic AMP would in fact be necessary for its chemotactic action.

It has been shown for another strain of *D. discoideum*, for the clone *v-12/M1*, that aggregation-competence in cell suspensions reaches a maximum between 8–10 h after end of growth, and then declines [17, 18]. Similarly, the activity of cell-bound phosphodiesterase declines in ax-2 cell suspensions after the 9-h period (Fig. 4). This parallelism suggests that, if the cell-bound phosphodiesterase participates in cyclic-AMP-receptor function, it also participates in the chemotactic response during cell aggregation.

It remains possible, however, that this suggestive correlation is misleading. Since bacteria also release cyclic AMP to which *D. discoideum* cells respond [19, 20], the increased sensitivity to cyclic AMP after shortage of bacteria would also provide a more efficient food-seeking mechanism. In this case, the correlation in time of aggregation-competence and high sensitivity to cyclic AMP would mean that the cells are simultaneously differentiated to realize two alternative developmental pathways: to aggregate and eventually build up a sporophore, or, if distant bacteria become available, to detect their weak chemotactic signals in order to continue growth.

*Note Added in Proof.* After submission of the manuscript, a paper by Pannbacker and Bravard [21] appeared on a cyclic AMP phosphodiesterase in particle fractions from *Dictyostelium discoideum* and the increased activity of this enzyme during aggregation. The function suggested by Pannbacker and Bravard for this phosphodiesterase, which appears to be identical with the enzyme studied in the present paper, is to supplement or replace the external phosphodiesterase in establishing concentration gradients of cyclic AMP.

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